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U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Vogtle Electric Generating Plant – Units 1&2
Licensee Event Report 2016-003-00
Tornado Missile Vulnerabilities Result in Condition Prohibited by Technical Specifications

Ladies and Gentlemen:

In accordance with the requirements of 10 CFR 50.73(a)(2)(i)(B), and 10 CFR 50.73(a)(2)(v)(D) Southern Nuclear Operating Company (SNC) is submitting the enclosed Licensee Event Report, 2016-003-00 for Units 1&2. This letter contains no NRC commitments. If you have any questions, please contact Dom Sutton at (706) 848-1428.

Respectfully submitted,

B. Keith Taber
Vice President – Plant Vogtle Units 1&2 - SNC

BKT/KCW

Enclosure: Units 1&2 Licensee Event Report 2016-003-00

cc: Southern Nuclear Operating Company
Mr. S. E. Kuczynski, Chairman, President & CEO
Mr. D. G. Bost, Executive Vice President & Chief Nuclear Officer
Mr. D. R. Madison, Vice President – Fleet Operations
Mr. M. D. Meier, Vice President – Regulatory Affairs
Mr. B. K. Taber, Vice President – Vogtle 1 & 2
Mr. B. J. Adams, Vice President – Engineering
Mr. D. D. Sutton, Regulatory Affairs Manager – Vogtle 1 & 2
RType: CVC7000

U. S. Nuclear Regulatory Commission
Ms. C. Haney, Regional Administrator
Mr. R. E. Martin, NRR Senior Project Manager – Vogtle 1 & 2
Mr. E. T. Coffman, Senior Resident Inspector – Vogtle 1 & 2

Vogtle Electric Generating Plant – Units 1&2
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Enclosure

Units 1&2 Licensee Event Report 2016-003-00

NRC FORM 366 (06-2016)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104		EXPIRES: 10/31/2018		
 LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block)		Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov , and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.						
1. FACILITY NAME Vogtle Electric Generating Plant (VEGP) Unit 1				2. DOCKET NUMBER 05000424		3. PAGE 1 OF 3		
4. TITLE Tornado Missile Vulnerabilities Result in Condition Prohibited by Technical Specifications								
5. EVENT DATE			6. LER NUMBER		7. REPORT DATE		8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR
10	26	2016	2016	- 003	- 00	12	19	2016
							FACILITY NAME Vogtle Unit 2	
							DOCKET NUMBER 05000425	
							FACILITY NAME	
							DOCKET NUMBER 05000	
9. OPERATING MODE		11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)						
1		☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)
		☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)
		☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)
		☐ 20.2203(a)(2)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)
10. POWER LEVEL		☐ 20.2203(a)(2)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)
		☐ 20.2203(a)(2)(iii)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)
		☐ 20.2203(a)(2)(iv)		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)
		☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(A)		☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(i)
		☐ 20.2203(a)(2)(vi)		☒ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)		☐ OTHER		Specify in Abstract below or in NRC Form 366A		
12. LICENSEE CONTACT FOR THIS LER								
LICENSEE CONTACT Dom Sutton, Regulatory Affairs Manager						TELEPHONE NUMBER (Include Area Code) 706-848-1428		
13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT								
CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	REPORTABLE TO EPIX
14. SUPPLEMENTAL REPORT EXPECTED						15. EXPECTED SUBMISSION DATE		MONTH
☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO								DAY
								YEAR
ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines) On October 26, 2016, Units 1 and 2 were in Mode 1 at 100 percent rated thermal power. Engineering personnel determined that the Turbine Driven Auxiliary Feedwater (TDAFW) Pump exhaust vent and the Condensate Storage Tank (CST) vents were not adequately protected from tornado generated missiles. Subsequent evaluations on November 29, 2016 resulted in the determination that the Atmospheric Relief Valves (ARVs) off the Main Steam Lines were also not adequately protected from tornado generated missiles. Operations subsequently declared the affected components inoperable, implemented Enforcement Guidance Memorandum (EGM) 15-002, "Enforcement Discretion for Tornado-Generated Missile Protection Noncompliance" and the required compensatory measures, and then declared the affected equipment operable but non-conforming. These tornado missile vulnerabilities existed since the original plant construction. Immediate compensatory measures include implementation of the severe weather procedures if a tornado is anticipated via a tornado watch or warning, notification to Maintenance and Operations departments that tooling is onsite for removal of these components should they become damaged from a tornado generated missile, and immediate inspection of these components following an event that could have generated missiles. Additional comprehensive corrective actions and permanent elimination of the vulnerabilities are planned in accordance with EGM 15-002.								

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

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		YEAR	SEQUENTIAL NUMBER	REV NO.
Vogtle Electric Generating Plant – Unit 1	05000-424	2016	- 003	- 00

NARRATIVE**A. Plant Operating Conditions Before the Event:**

Unit 1, Mode 1, 100 percent power
Unit 2, Mode 1, 100 percent power

No plant transients are associated with this event. Thus, no Structures, Systems, or Components (SSCs) were inoperable at the start of this event which contributed to this event.

Background NRC Documents:

Enforcement Guidance Memorandum (EGM) 15-002, "Enforcement Discretion for Tornado-Generated Missile Protection Noncompliance," provides guidance to exercise enforcement discretion when an operating power reactor licensee does not comply with a plant's current site-specific licensing basis for tornado-generated missile protection. Specifically, discretion would apply to the applicable Technical Specification (TS) Limiting Condition(s) for Operation (LCO) that would require a reactor shutdown or mode change in the event a licensee could not meet TS LCO required action(s) within the TS completion time.

Interim Staff Guidance DSS-ISG-2016-01, "Clarification of Licensee Actions in Receipt of Enforcement Discretion Per Enforcement Guidance Memorandum EGM 15-002," provides interim staff guidance to facilitate staff understanding of expectations for consistent oversight associated with implementing enforcement discretion for tornado missile protection noncompliance(s) per EGM 15-002.

Appendix A to DSS-ISG-2016-01 provides guidance for acceptable initial and comprehensive compensatory measures for licensee use in implementing the enforcement discretion outlined in EGM 15-002. The licensee should declare (log) the utilization of EGM 15-002, inform the resident inspector, and enter the issue into the corrective action program. For initial compensatory measures, it is expected that the measures listed are already in place at sites that may be affected by severe weather, such as tornados and/or hurricane force winds. The measures should be verified as current and readily deployable within a very short timeframe.

B. Description of the Event:

On October 26, 2016, during the evaluation of protection of TS equipment from damage from tornado generated missiles, Vogtle Site Engineering identified nonconforming conditions in the plant design such that the specified TS equipment did not meet the current design basis for protection against tornado generated missile impacts. Three systems contained equipment that were affected to the extent that an impact on operability was credible. The TDAFW Pump could be rendered inoperable due to a tornado generated missile strike on the exhaust vent piping that extends outside of the pump house. The CST vents could be rendered inoperable due to a tornado generated missile strike, as well as the ARVs which extend above the roof of the main steam valve rooms. These SSCs were declared inoperable and EGM 15-002 was invoked. Compensatory measures were implemented within the allowed time by the applicable LCOs and the associated systems were declared operable but nonconforming.

This condition is being reported per the following criteria:

- 10 CFR 50.73(a)(2)(i)(B) for any condition that is prohibited by TS.
- 10 CFR 50.73(a)(2)(v)(D) for any condition that could have prevented the fulfillment of a safety function of structures or systems needed to mitigate the consequences of an accident

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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C. Cause of the Event:

Due to the Historical nature of this vulnerability, a specific cause was not able to be identified.

D. Safety Consequences:

As documented in EGM 15-002, tornado missile scenarios that may lead to core damage are very low probability events because safety-related SSCs are typically designed to withstand effects of tornados. For a tornado missile-induced scenario to occur, a tornado would have to hit the site and result in the generation of missiles that would hit and fail vulnerable, unprotected safety-related equipment, and/or unprotected safety-related subcomponents in a manner that is non-repairable and non-recoverable. In addition, because plants are designed with redundancy and diversity, the tornado missiles would have to affect multiple trains of safety systems and/or means of achieving safe shutdown.

The NRC has completed a generic risk analysis of potential tornado missile protection noncompliances to examine the risk significance of these scenarios. This assessment documents a conservative, bounding-type analysis of the risk significance for plant facilities. The generic analysis assumed that core damage would occur if a tornado hit a plant located in the most active tornado region in the country and that it caused a tornado-generated missile to fail all emergency core cooling equipment at the plant with no ability to recover. Given this conservative assumption, it has been determined that the core damage frequency (CDF) associated with tornado missile-related noncompliances are well below CDFs requiring immediate regulatory action. In summary, the generic bounding risk analysis performed by the NRC concluded that this issue is of low risk significance.

During a postulated design basis tornado, the conditions documented could have resulted in a loss of function for the TDAFW pump, CSTs, or the ARVs. All these systems are used to mitigate the effects of postulated accidents by providing an alternate path for core heat removal. No actual safety consequence occurred, as Vogtle did not experience an actual tornado missile event. Therefore, enforcement discretion until June 10, 2018, will not impose significant additional risk to public health and safety.